DBMS Lab Mini Project – Guideline & Instruction

(Complex Engineering Problem)

Department of CSE

Course: Database Management Systems Lab
Project Type: Mini Project (Group-based)
Total Marks: 40

Targeted Attributes for DBMS-Complex Engineering Problem:

Attribute	Description
EP1	Ability to apply knowledge of mathematics, science, engineering fundamentals and engineering specialization to the solution of complex engineering problems.
EP2	Ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using principles of mathematics, natural sciences and engineering sciences.
EP4	Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Instructions to All Student Groups:

1. Group Distribution & Assigned Topics

Group No.	Project Title	Problem Domain
Group 1	University Academic Management System	Education ERP
Group 2	Hospital Appointment & Report System	Healthcare
Group 3	Job Recruitment Portal	Employment Technology
Group 4	Smart Library with Reservation System	Knowledge Resource
Group 5	Public Event Ticketing & Feedback System	Event & Entertainment

Each group will tackle a **real-world problem** involving multiple entities, large-scale data operations, and advanced database features. You must follow a structured process starting from problem identification to SQL implementation and testing.

Project Components & Submission Structure

A. Problem Identification & Scope (EP2) – [5 Marks]

- Clearly define the problem.
- Identify stakeholders and system users.
- Discuss complexity: constraints, multi-entity dependencies, and performance needs.

B. System Design (EP1) – [5 Marks]

- Create a detailed **ER diagram** with relationship cardinalities.
- Normalize your schema to at least **3NF**.
- Provide full `CREATE TABLE` scripts with constraints (`PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, etc.).

C. SQL Implementation & Complex Queries (EP1, EP2) – [10 Marks]

- Populate with realistic data (at least 20 records per table).
- Write **10+ SQL queries**, covering:
  - Basic SELECT with WHERE, GROUP BY, HAVING
  - JOINS (INNER, LEFT, RIGHT)
  - NESTED QUERIES and SUBQUERIES
  - VIEWS, TRIGGERS, TRANSACTIONS, STORED PROCEDURES

D. Complex System Functions & Automation (EP3) – [5 Marks]

- Demonstrate system complexity through:
  - Business logic in `Triggers` or `Procedures`
  - Data integrity with constraints and transactions
  - System roles and privileges (optional but appreciated)

E. Report & Reflection (All EPs) – [5 Marks]

- A PDF report (max 6 pages) including:
  - Introduction & Problem Definition
  - ER Diagram & Schema
  - SQL Code Snippets with Sample Output
  - Reflection: how each EP was addressed

Deliverables:

You must submit the following:

- ✔ MySQL Workbench file (`.mwb`)
- ✔ SQL Script file (`.sql`)
- ✔ PDF Report (max 6 pages)
- ✔ Screenshot Folder (query results/output)
- ✔ [Optional] 5-minute group presentation

Timeline:

- **Report Submission Deadline:** 15.08.2025
- **Presentation & Viva (Optional):** 18.08.2025

Evaluation Rubric (Total 40 Marks)

Component	Description	Attribute	Marks
A. Problem Identification & Scope	Define the problem, identify stakeholders, constraints, and real-world complexity.	EP2	5
B. System Design & Schema	ER diagram, normalization (up to 3NF), and table creation scripts with constraints.	EP1	5
C. SQL Implementation	10+ SQL queries with joins, subqueries, views, triggers, procedures, and transactions.	EP1, EP2	10
D. Investigation & Analysis	Conduct exploratory analysis, test hypotheses using SQL queries, and present analytical findings with interpretation.	EP4	5
E. Final Report	Include all components, outputs, screenshots, and a summary explaining how EP1, EP2, and EP4 were achieved.	All	5
F.Viva	Based on overall project and project report	All	10
Total			**40**

Reminder:

Your work must reflect **team collaboration**, **real-world complexity**, and a deep **understanding of engineering design principles**. Plagiarism or template-based designs will not be accepted. Each group must tackle **unique data logic** and ensure **traceability to EP1–EP3** outcomes.

Appendix

How we Can Address EP1, EP2, and EP4 through the DBMS Mini Project?

EP1 – Apply Knowledge of Engineering Fundamentals

Goal: Apply your theoretical knowledge from DBMS, data modeling, SQL, and normalization in a practical, real-world project.

How to address:

- Use your understanding of **entity-relationship modeling** to design the database.
- Apply **normalization rules (1NF to 3NF)** to remove redundancy and ensure consistency.
- Write **optimized SQL queries** using SELECT, JOIN, GROUP BY, etc., to solve real tasks like generating reports, managing transactions, or creating analytics.
- Implement **constraints, primary/foreign keys** to ensure referential integrity.

Example: Create a normalized database schema for a hospital that ensures all appointments, reports, and patient details are logically structured and interrelated.

EP2 – Identify, Formulate & Analyze Complex Problems

Goal: Demonstrate that you can identify a real-world engineering problem and break it down into smaller solvable components.

How to address:

- Begin your report by **clearly defining the problem statement**.
- Identify **key entities, relationships, and processes** involved (e.g., patient registration, doctor assignment, billing, etc.).
- Formulate **data requirements**, constraints, and potential bottlenecks.
- Analyze the complexity using **ER diagrams and logical relationships** between data elements.

Example: In a job portal, analyze how job seeker data connects with applications, interview schedules, and employer feedback, and design the schema accordingly.

EP4 – Conduct Investigations Using Research-Based Methods

Goal: Use your database to **investigate** a real-world question using research-oriented techniques like data grouping, pattern analysis, and summarization.

How to address:

- Formulate **one or two investigative questions** related to your project (e.g., Which doctor has the highest appointment load? Which event got the best user rating?).
- Use **SQL queries** (with `GROUP BY`, `HAVING`, `AVG`, `COUNT`, etc.) to **explore patterns, frequencies, or trends** in your dataset.
- Interpret the findings in your report using **tables, summary text, or simple visualizations**.

Example: For a library project, investigate which categories of books are borrowed the most and what days students prefer to borrow books.